Project Plan
Product-based Capstone Project

Team ID : **C241-PS263**

Team Member : (Pelase adjust according to your team members)

1. (ML) m004d4ky3162 – Rianco Marcellino Andreas – ITS University - [Active/Inactive]
2. (ML) m002d4ky1567 – Amjad Adhie Prasetyo – Bandung Institute of Technology - [Active/Inactive]
3. (ML) m001d4ky1742 – Rafli Radithya – University - [Active/Inactive]
4. (CC) c002d4kx0108 – Aditya Inas Hamidah – Bandung Institute of Technology - [Active/Inactive]
5. (CC) c004d4ky0464 – Muhammad Naufal Rafianto – ITS University - [Active/Inactive]
6. (MD) a004d4ky4156 – Muhammad Daffa' Fisabilillah – ITS University - [Active/Inactive]
7. (MD) a009d4ky4572 – Laurens Fernando Pasaribu – Gunadarma university - [Active/Inactive]

Final Selected Themes:

Economics Empowerment: Navigating Sustainable Economies for All

Title of the Project:

TerraVision

Executive Summary/Abstract:

In recent years, real estate investment has surged in popularity due to housing demands, economic expansion, and passive income opportunities. However, entering this complex market poses challenges, particularly for newcomers lacking experience and insight. Challenges include understanding market dynamics, accurate property valuation, and effective asset management post-purchase. Novices often struggle to assess property profitability, leading to potential financial risks.

In response, TerraVision introduces a comprehensive platform, offering simulation and real estate price prediction tools to assist beginners. TerraVision is set to redefine real estate investment with its groundbreaking platform. With a keen understanding of the burgeoning interest in real estate investment, propelled by housing demands, economic expansion, and passive income prospects, TerraVision aims to address the hurdles encountered by newcomers venturing into this intricate realm. Surveys conducted by our team underscore

Project Plan
Product-based Capstone Project

a prevalent eagerness among diverse demographics to explore real estate investment, hampered by a lack of foundational knowledge.

The primary goal of TerraVision's platform is two-fold: to streamline the learning curve for aspiring real estate investors and to furnish a comprehensive toolset for scrutinizing and juxtaposing potential returns on diverse properties before committing to investments. TerraVision distinguishes itself with its cutting-edge AI-driven prediction algorithms, furnishing lifelike investment simulations, tailored analyses, exhaustive property ratings, nominal-based property filtering, and an adaptable design. Employing the Agile Scrum methodology, TerraVision emphasizes agile principles, team collaboration, iterative development, and incremental advancement.

How did your team come up with this project?
Data collection involved surveys via Google Forms and literature review targeting diverse demographics, from children to the elderly, to gauge public understanding of real estate. Literature review encompassed articles, academic works, and books. Development utilized Agile Scrum methodology to produce a functional website, followed by testing to assess performance.

Agile Scrum was chosen for its effectiveness in developing applications, emphasizing teamwork, iterative development, and collaboration. Key roles include the Product Owner, responsible for communicating vision and priorities; the Scrum Master, facilitating communication; and the Development Team, responsible for project implementation.

Project Scope & Deliverables:

Task	Responsible Team Member(s)	Timeline
1. Requirement Analysis	**All**	**Week 1**
- Gathering Team and Team requirements and expectations / All team members participate in gathering team and project requirements and formulating product plans.	All	Week 1

bangkit
Google goto tokopedia traveloka

Kampus Merdeka
INDONESIA JAYA

Project Plan
Product-based Capstone Project

- Initial discussions about the product plan were conducted by all team members to generate ideas and division of labor.	All	Week 1
2. Infrastructure Setup	All	Week 1
- The entire team collaborates to provision cloud servers and databases	All	Week 2
- implementation of security measures carried out by Cloud Computing	Cloud Computing	Week 2
- The Cloud Computing Team is responsible for ensuring system scalability and performance.	Cloud Computing Team	Week 2
3. Frontend Development	Mobile Development	Week 3
- The Mobile Development team is responsible for designing UI/UX for web and mobile interfaces.	Mobile Development Team	Week 3
- Frontend development for web applications is done by MD.	Mobile Development 1	Week 3
- Build mobile applications (Android & iOS)	Mobile Development 2	Week 3-4
4. Backend Development	Cloud Computing	Week 3
- Set up backend architecture and APIs	Cloud Computing Team	Week 3
- The implementation of data storage and data retrieval is done by the Cloud Computing team.	Cloud Computing 1	Week 2
- Algorithm development for real estate prediction is done by the Machine Learning Team.	Machine Learning Team	Week 3-4

Project Plan
Product-based Capstone Project

5. Machine Learning Models	Machine Learning	Week 4-5
- The Machine Learning team is responsible for collecting and processing real estate data.	Machine Learning Team	Week 4
- Training of predictive models for investment value is done by the Machine Learning team.	Machine Learning Team	Week 4-5
- Validation and refinement of the ML model is done by the Machine Learning team.	Machine Learning Team	Week 5
6. Integration & Testing	All	Week 6
- All team members work together to integrate the frontend with the backend.	All	Week 6
- Functionality and user experience testing is done by all team members.	All	Week 6
- Debug	All	Week 6-7
7. Deployment & Launch	All	Week 7-8
- All team members were involved in the implementation and launch of the web and mobile applications.	All	Week 7
- Performance monitoring and user feedback are conducted by the entire team.	All	Week 7
- Handling post-launch issues is also a shared responsibility of the team.	All	Week 8

Project Plan
Product-based Capstone Project

Project Schedule:

Milestone	Description	Timeline
Milestone-0 [All Members] - Kick-off meeting - Requirements Gathering - Composing Project Plan	Our team conducts a kick-off meeting to plan the overall project including finding the problem statement we can solve using mobile app, what tech stack we will use, etc. We also compose the project plan as initial reference for the project.	29 April - 13 Mei 2024
Milestone-1 [Cloud Computing] Making ERD for Database Modelling and Designing Architecture [Mobile Development] Designing UI / UX for Application [Machine Learning] Exploring and collect potential data from various resources & performing data pre-processing	Cloud computing team making ERD diagram that later be translated into the application database model filled by the data required for the application. Mobile development team designing UI/UX for the application that supports its functionality, attracting users, and ensuring users usability & comfort while using the app. Machine learning team collects data related to real estate price e.g; location, housing area, etc and performs data pre-processing like data cleaning.	14 Mei - 23 Mei 2024

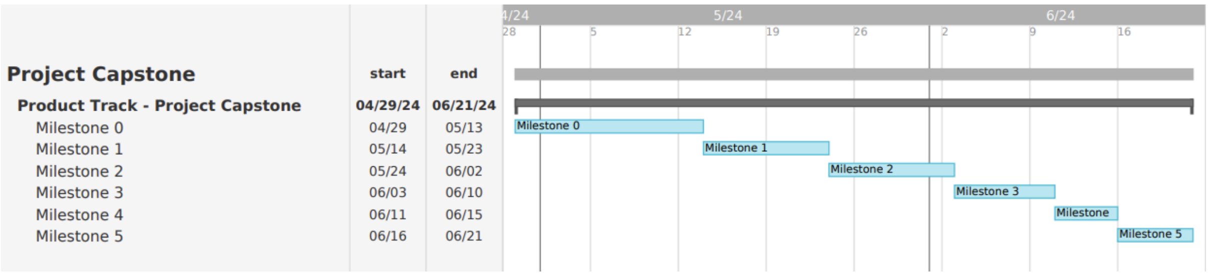
Project Plan
Product-based Capstone Project

Milestone-2 [Machine Learning] Creating model for real estate price prediction [Cloud Computing] Set up database and data storage to store data, developing App back-end [Mobile Development] Developing App Front-End	Machine learning team creating prediction model from data collected Cloud computing team setting-up overall architecture of the cloud system, creating backend for the mobile app, storing machine learning model and connecting the application with machine learning service. Mobile development developing app front-end based on the design prototyping done before	24 Mei - 2 Juni 2024
Milestone-3 [Cloud Computing] Testing Mobile App Back-End [Mobile Development] Integrating the mobile app with cloud computing backend	Cloud computing team testing back-end using tools like postman continued by Mobile Development team integrating it with the front-end.	3 Juni - 10 Juni 2024
Milestone-4 [Mobile Development] Perform unit testing for mobile app [All Members] Fixing bugs	Mobile development team perform unit testing for application while cloud computing and machine learning team find possible bugs in the application.	11 Juni - 15 Juni 2024
Milestone-5 [Cloud Computing] Deploying to Production Environment	Cloud Computing team deploying the mobile app into production environment and monitor its performance	16 Juni - 21 Juni 2024

Project Plan
Product-based Capstone Project

[All Members] Create Project Documentation	All members of the team creating project documentation and pitch deck for the deliverables	

Here is the project Gantt Chart based on above timeline:



Based on your team's knowledge, what tools/IDE/Library and resources that your team will use to solve the problem?

- Google Colab:
- Jupyter Notebook / JupyterLab
- Visual Studio Code (VS Code)
- Android studio
- ML Deployment (TensorFlow.js, TFLite, or similar alternatives)
- Postman
- Google Cloud Platform (App Engine, Cloud Storage, Cloud SQL, and Cloud Build)
- Firebase
- GitHub & Git

Based on your knowledge and explorations, what will your team need support for?

- Mentors who can help us decide which tools or resources are most suitable to use and guide us in completing the project
- Datasets for the recommendation system include location, size, image and owner of the property

Based on your knowledge and explorations, tell us the Machine Learning Part of your Capstone!

We will each train a TensorFlow machine learning model to predict house prices in the Jabodetabek area for one or several years in advance. The model will be built using 80% of the available data for training, and then validated using 10% of the remaining data. We will select the model with the highest accuracy without overfitting, and test it again using the remaining test data. The output of the model will be the predicted house prices for the given year, along with time series graphs to show the fluctuations.

Based on your knowledge and explorations, tell us the Mobile Development Part of your capstone?

We are planning to develop an Android application using Kotlin and Android Studio. This application will be integrated with a backend API, previously configured by our Cloud/Backend team, to present real estate data. This includes property locations, images, dimensions, current prices, previous prices, and ownership status. Furthermore, the application will leverage machine learning technology by incorporating a TensorFlow Lite model developed by our Machine Learning team. This model is designed to predict future property price trends and provide strategic recommendations to users. We will also optimize the application to support various screen sizes, ensuring a superior user experience.

Link Figma :
https://www.figma.com/file/8hiSJK4HD2t0qtHi1Q2Tvy/Untitled?type=design&node-id=0%3A1&mode=design&t=yayxtUHb7mqbdjOt-1

Based on your knowledge and explorations, tell us the Cloud/Web/Frontend/Backend Part of your capstone?

For the Capstone Project Google Cloud/Backend part is to leverage GCP services like Compute Engine, Cloud Functions, and Cloud SQL for infrastructure and API development. Utilize AI Platform for machine learning tasks and Cloud Monitoring for performance tracking. Ensure security with IAM and Firebase Authentication for API access.

Based on your team's planning, is there any identifiable potential Risk or Issue related to your project?

There are several aspects that need to be clarified and expanded upon :

1) Complexity of Variables: AI may have limitations in accounting for all relevant variables that affect property value. For instance, besides external factors such as natural disasters, there are also internal factors influencing property value such as building physical conditions, surrounding infrastructure, and local regulatory policies. A lack of deep understanding of these factors can result in uncertainty in AI prediction outcomes.
2) Dependency on Historical Data: AI models tend to rely on historical data to make predictions. However, real estate market conditions can change rapidly and may not

always reflect past trends. Therefore, there is a risk that AI may not capture significant market changes quickly enough, especially if there is no actual data reflecting current conditions.

3) Limitations in AI Adaptation Capability: Although AI can learn from new data, there are limitations on how quickly and how well AI can adapt to drastic market condition changes. For example, if significant regulatory changes or political events affect the real estate market, AI may need time to adjust its model or may even be unable to accommodate these changes effectively.

Any other notes/remarks we should consider on your team's application

In developing TerraVision, it's imperative to prioritize user experience by ensuring intuitive interface design and seamless functionality across devices. Additionally, fostering a robust community engagement platform where users can share insights, tips, and experiences can enhance user satisfaction and retention. Regular updates and enhancements based on user feedback and market trends will further solidify TerraVision's position as a leading platform for real estate investment exploration and analysis.